

Teamwork Agreement Template

For the MathWorks Classroom Challenge Projects

Project: Drone Payload Capacity and Structural Design Analysis

Team Name: ____JOOP Drone Makers (Team 6)_____

Instructor: ____Kevin Graham_____

Program/Class: ____EPP Drone Project_____

Members and Roles

1. Ian Mckillop: Project Manager_____
2. Joseph Hua: Analysis/Validation Lead_____
3. _____
4. _____
5. _____

1) Purpose & Scope

We agree to collaborate to deliver a high-quality, ethical, and reproducible project. Our work will follow the project guidance and produce the specified artifacts (e.g. plots, code, and visualizations).

2) Roles & Responsibilities

- Project Manager (PM): Schedules meetings; maintains timeline; ensures tasks are tracked and disagreements are resolved.
- Modeling Lead: Owns core MATLAB models/simulations
- Analysis/Validation Lead: Performs hand calculations and cross-checks simulation outputs
- Documentation & Visualization Lead: Compiles report (with figures/tables), prepares slides (if needed), and builds visual artifacts (e.g. animations or plots)
- Quality Assurance & Reproducibility Lead: Enforces coherent style across artifacts, ensures work is not overwritten across different versions, ensures data quality, tests/re-runs codes to ensure robustness and reproducibility

3) Communication Norms

- Primary channel: ___Discord___ (e.g. Slack)
- Secondary channel: ___Email___ (e.g. E-mail for summaries/decisions)
- Response time: ___24 hours___ on weekdays and ___48 hours___ on weekends (e.g. within 24 hours on weekdays; within 48 hours on weekends)
- Meeting cadence: ___Wednesdays via Discord___ (e.g. weekly working meeting (60–90 minutes on Tuesdays via Zoom))
- Decision records: ___Discord___ (e.g. PM logs major decisions in a running document)

4) Collaboration & Tools

- Version control: ___Github Repo___ (e.g. one shared GitHub repo or Google Drive folder)

- Data management: All datasets, parameters, and assumptions documented in _____README.md_____ (e.g. README.md)

File naming & style: _____DTeam6_____ (e.g. Snake_case, descriptive names)

- Backups: _____BackupDTeam6_____ (e.g. PM tags old versions according to numbered system)

5) Quality Standards (Definition of 'Done')

Requirements meet project description, code is reproducible (runs from the README.md file, all specified figures regenerate, at least one teammate has tested edge cases, all assumptions, units, and references are documented in the report)

6) Timeline & Milestones

- Week 1: _____Understanding the project, setting up communication, assigning roles, beginning MATLAB code, and prototype._____
- Week 2: _____Starting CAD, continuing MATLAB, completing a first draft of the project._____
- Week 3: _____Optimizing project, Testing for flaws/issues_____

- Week 4: _____ Work on visuals (slides, report), presentation practice, final tweaks and adjustments to project. _____

(e.g.

Week 1: Problem framing, literature/background review, dataset/parameters,

Week 2: First pass models (hand calculations + baseline MATLAB scripts); initial plots/visuals.

Week 3: Validation and quality assurance

Week 4: Report, slides, presentation rehearsal, repository cleanup)

7) Decision-Making & Conflict Resolution

- Default: _____ Discuss to work things out _____ (e.g. consensus after discussion; PM facilitates)
- If blocked >48 hrs: _____ Majority vote _____ (e.g. Time-limited debate, majority vote, PM documents decision)
- Technical disagreements: _____ Evidence-based comparison before vote _____
- Escalation: If unresolved, consult instructor/mentor with a concise memo.

8) Workload, Accountability & Attendance

- Task board: _____ Assign tasks through Discord/Google Document and keep each other updated on status _____ (e.g. To-Do / Doing / Review / Done with owners and due dates documented in README.md)
- Accountability: _____ Explain whatever you are taking accountability for and provide a new deadline to complete it _____ (e.g. Missed deadline → recovery plan within 24 hrs; repeated misses trigger role redistribution.)
- Attendance: _____ Let group know ahead of time you're going to be absent and catch up asynchronously. _____ (e.g. If you'll miss a meeting, give 24-hour notice and share updates asynchronously.)

10) Accessibility & Inclusion

We commit to respectful collaboration, equitable turn-taking, and flexibility for documented accessibility needs.

11) Deliverables Checklist

- At least two drone design sketches showing key geometric features
- Completed MATLAB script + CAD of drone design(s)
- Results of thrust-to-weight analysis are presented either in a summary table or a plot
- Results of finite element analysis
 - Numerical values presented in a summary table
 - Visualizations of x-, y-, and z-displacement and Von Mises stress
- A final design recommendation justified using the results of your analyses

□ _____

□ _____

□ _____

(e.g. Code repository with clear structure and README, hand calculations linked to modeled results for cross-validation, specific figures & visualizations, final report with specified sections)

12) Agreement & Signatures

Name & Signature: _____ Joseph Hua _____ Date: ____7/8/2026____

Name & Signature: _____ Ian Mckillop _____ Date: ____7/8/2026____

Name & Signature: _____ Thura Zin _____ Date: ____7/8/2026____

Name & Signature: _____ Date:
